

CIRCUIT ARCHITECTURE REVIEW

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I. Introduction

This document presents a review of the architecture of a comparator. The study focuses on describing the static and dynamic characteristics that define its performance, as well as evaluating the practical limitations derived from non-ideal effects such as input offset, noise, and propagation delay.

It also reviews the topology selected for the comparator.

A. Design Specifications

The following table shows the Electrical specifications for the comparatorCircuit.

Table 1. Electrical Specifications for the comparator Circuit. TBS == to be simulated. TBD == to be defined.

N°	Symbol	Parameter	Unit	Target		
				Min	Typ	Max
1	VDD	Supply Voltage	[V]	1.62	1.8	1.98
2	Ivdd	Current Consumption	[uA]			TBD
3	Fpwm	Switching Frequency	[MHz]	1.8	2.00	2.20
4	Drange	Duty Range	[%]		TBS	
5	Vctrl	Control Voltage	[V]		0.8	
6	CL	Load Capacitance	[fF]		TBD	
7	Ao	Open loop gain	[dB]		TBD	
8	BW	Band width	[MHz]		TBD	

II. State of the Art

A. Characterization of a Comparator

To illustrate the characteristics of a comparator, the amplifier shown in Figure 1 will be considered, where V_P is the positive input of the comparator. Applying a positive voltage to V_P will cause the comparator's output to become positive. V_N is the negative input of the comparator, and applying a positive voltage to V_N will cause the comparator's output to become negative. The upper and lower voltage limits of the comparator's output are defined as VOH and VOL , respectively [1].



a) Circuit symbol for a comparator

b) Ideal transfer curve of a comparator

Fig 1. Circuit and Ideal transfer curve of a comparator. [1]

Static Characteristics

An ideal comparator was previously defined. Although this type of behavior is impossible in a real situation, it can be modeled using ideal circuit elements with mathematical descriptions, which allow us to understand the characteristics of a comparator in a simpler way. Figure 2 shows a model of this ideal circuit.

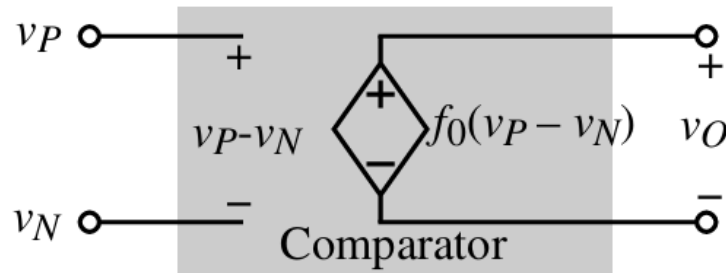


Fig 2. Model for an ideal comparator. [1]

Analyzing Figure 1.a, the comparator output is high (V_{OH}) when the difference between the non-inverting and inverting inputs is positive, and low (V_{OL}) when this difference is negative. The output changes state in response to a change in the input voltage ΔV , where ΔV tends toward zero.

This implies an infinite gain, as shown in the following equation.

$$A_v = \lim_{\Delta V \rightarrow 0} \frac{V_{OH} - V_{OL}}{\Delta V}$$

Once the operation of an ideal comparator has been reviewed, we analyze the behavior of a first-order model. Figure 3 shows the DC transfer curve of this model.

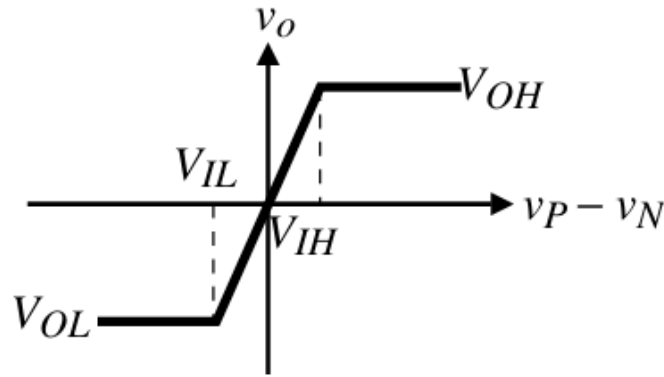


Fig 3. Transfer curve of a comparator with finite gain. [1]

where V_{IH} and V_{IL} represent the input voltage difference required to obtain an output at its upper and lower limits, respectively. The gain of this model can be expressed as follows

$$A_v = \frac{V_{OH} - V_{OL}}{V_{IH} - V_{IL}}$$

this feature is very important because it describes how the comparator works and defines the minimum amount of input change (resolution) required for the output to oscillate between the two binary states.

A non-ideal effect is the input-offset voltage V_{OS} . This effect would not be an issue if it were predictable; however, it varies randomly from one circuit to another for a given design. Another non-ideal effect to consider is noise. The noise of a comparator is modeled in the transition region, generating uncertainty in that zone. Although a comparator is not designed to operate in the transition region, this noise becomes relevant in circuits where the comparator is employed.

Dynamic Characteristics

The dynamic characteristics of the comparator define how the comparator responds over time to a differential input signal. These characteristics include both small-signal and large-signal behavior [1].

The comparator has a propagation delay between the differential input signal and the output transition. This is a critical parameter that limits the conversion speed in an A/D converter. This delay depends on the input amplitude: the greater the amplitude, the shorter the delay, until a limit defined by the slew rate (large-signal behavior) is reached.

The small-signal behavior is characterized by the frequency response of the comparator. Where it can be modeled as a differential voltage gain A_v , which is expressed as

$$A_v(s) = \frac{A_v(0)}{\frac{s}{\omega_c} + 1} = \frac{A_v(0)}{s\tau_c + 1}$$

The propagation delay of a comparator can be described by an exponential response when applying a differential input step. Figure 4 shows this behavior.

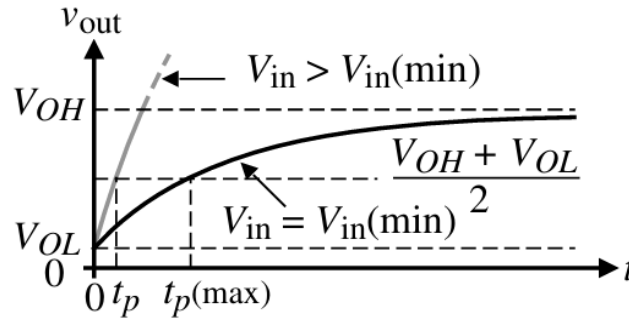


Fig 4. Small-signal transient response for a comparator. [1]

If V_{in} is larger than $V_{in(min)}$ (minimum voltage change at the comparator input.), the output rise (or fall) time is faster. Therefore it can be said that by applying a $V_{in(min)}$ input we will obtain a maximum propagation delay time.

When an overdrive is applied to the input which makes V_{in} greater than $V_{in(min)}$, the propagation delay time depends on the amplitude of the applied signal: the greater the input overdrive, the shorter the propagation time. However, when the overexcitation is sufficiently large, the comparator stops behaving in small-signal mode and enters large-signal mode. In this condition, the limitation is no longer imposed by the propagation delay time, but by the slew rate of the circuit. This means that the propagation time is governed by the maximum charge or discharge speed of the internal capacitances. Therefore, improving the sink or source current capability of the comparator is key to reducing the propagation time in this regime.

III. Topology Selection

A. Review the topology of the proposed solution.

A review of the literature on the three-stage high-speed comparator topology was carried out, since this was the previously chosen optimal solution topology.

This comparator consists of three stages. The first is the pre-amplification stage, which amplifies the input signal to improve the comparator's sensitivity and isolates the input from the switching noise from the comparator's positive feedback stage. The positive feedback stage determines which of the input signals is larger. The output buffer amplifies this information and outputs a digital signal [2].

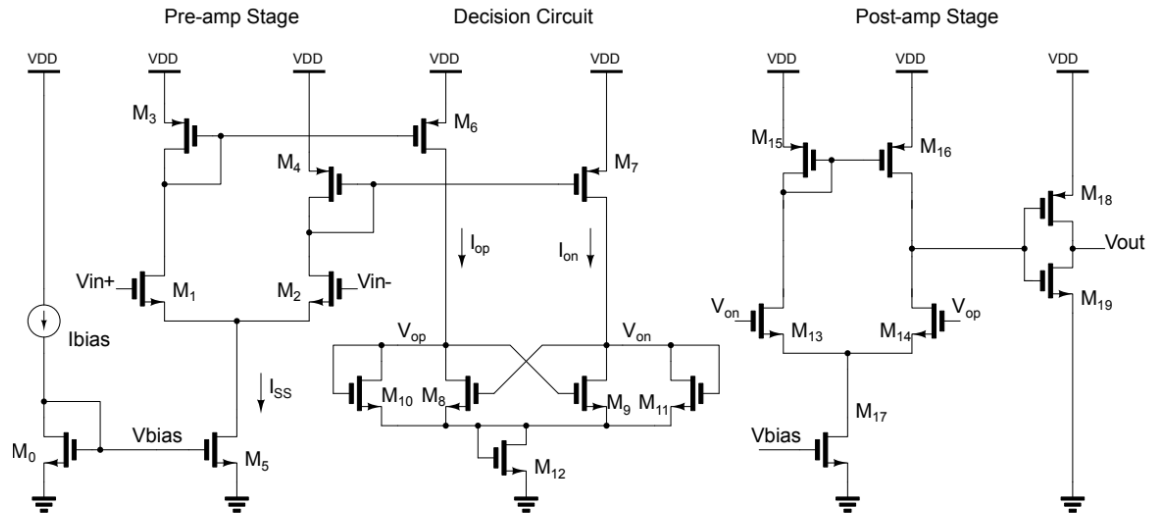


Fig 5. Comparator topology. [3]

Pre-amp Stage:

The circuit of Figure 6 is a differential amplifier with active loads. The sizes of M1 and M2 are set by considering the diff-amp transconductance, g_m , and the input capacitance. The transconductance sets the gain of the stage, while the input capacitance of the comparator is determined by the sizes of M1 and M2 [2].

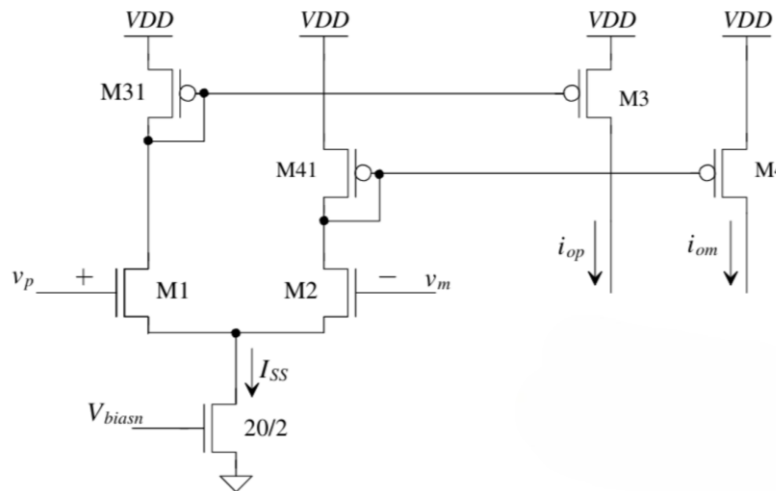


Fig 6. Preamplification stage. [2]

Note that there are no high-impedance nodes in this circuit, other than the input and output nodes. This reduces slow internal time constants and allows for fast switching. Furthermore, using active loads (PMOS current mirror) allows for relatively high r_{out} gains without introducing large physical resistors (which would be slower and less compact).

Now we can relate the small-signal output currents i_{op} and i_{om} with the input voltages by

$$i_{op} = \frac{gm}{2}(v_p - v_m) + \frac{I_{SS}}{2} = I_{SS} - i_{om}$$

Decision

The decision circuit is the heart of the comparator and should be capable of discriminating between mV-level signals.

Circuit:

Figure 7 shows the circuit used in the comparator under development, which uses positive feedback through the cross-connection of gates M6 and M7, which allows increasing the gain of the decision element. In addition, it is desirable to incorporate hysteresis into the design in order to suppress the effect of noise at the input.

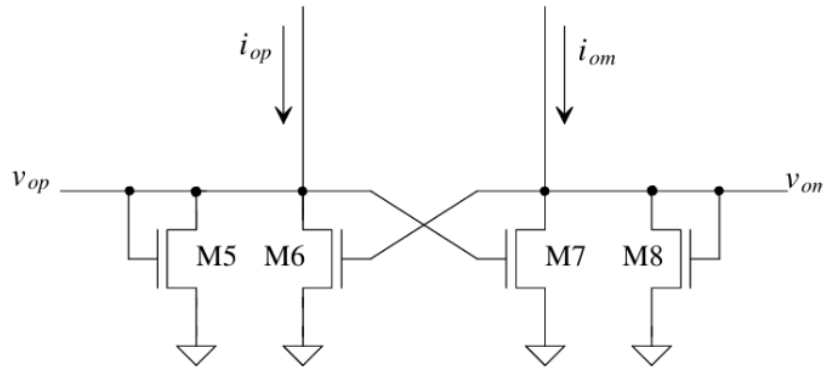


Fig 7. Positive feedback decision circuit. [2]

Assuming that i_{op} is much larger than i_{om} so that M5 and M7 are on and M6 and M8 are off. We also assume that $\beta_5 = \beta_8 = \beta_A$ and $\beta_6 = \beta_7 = \beta_B$, so that, v_{om} is approximately 0 [V] and v_{op} is

$$v_{op} = \sqrt{\frac{2i_{op}}{\beta_A}} + V_{THN}$$

The currents flowing in M7 is

$$i_{om} = \frac{\beta_B}{2}(v_{op} - V_{THN})^2$$

And the current flowing in M5 is

$$i_{op} = \frac{\beta_A}{2}(v_{op} - V_{THN})^2$$

Note that M5 and M7 are a current mirror, we can write that

$$i_{op} = \frac{\beta_A}{\beta_B} \cdot i_{om}$$

If $\beta_A = \beta_B$ the switching takes place when $i_{op} = i_{om}$. The differences in β_s cause hysteresis in the comparator.

The switching voltages can be obtained by performing the following equation

$$VSPH = vp - vm = \frac{I_{SS}}{gm} \cdot \frac{\frac{\beta_A}{\beta_B} - 1}{\frac{\beta_A}{\beta_B} + 1} \text{ for } \beta_B \geq \beta_A$$

$$VSPH = -VSPH$$

Now, as the V_{GS} of M8 increases, exceeding V_{THN} , M6 draws current from M5, which decreases the source-drain voltage of M5 and M6, causing M7 to turn off.

Post-amp

Stage:

The main purpose of the output buffer is to convert the output of the decision circuit into a logic signal. The output buffer should accept a differential input signal and not have slew-rate limitations.

References

- [1] D.R.H. Phillip E. ALLEN. CMOS Analog Circuit Design. OXFORD UNIVERSITY PRESS, 2011.
- [2] R. Jacob BAKER, Harry W. LI, and David E. BOYCE. CMOS. Circuit design layout and simulation. Fourth Edition. IEEE Press, 2019.
- [3] O. J. Amaya Amaya and V. H. Muñoz López, "Design of Base-Band Circuits for a PWM Controller in a DC-DC Buck Converter in CMOS Technology," B.Sc. Thesis, Escuela de Ingeniería Eléctrica, Electrónica y de Telecomunicaciones, Universidad Industrial de Santander, Bucaramanga, Colombia, 2024.